

George Thanasoulis

ELECTRICAL & COMPUTER ENGINEER

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LinkedIn Profile: George Thanasoulis

Fulfilled Military Obligation 2023/2024.

GitHub Profile: <https://github.com/GeorgeThan414>

Electrical and Computer Engineer and MSc candidate in Artificial Intelligence at Aristotle University of Thessaloniki, currently working as a Data Scientist at CERTH on two active European research projects using Machine Learning and Generative AI. Experienced across full Machine Learning lifecycle, from data mining and model development to cloud-native deployment and integration between system components. Seeking full-time opportunities to apply Artificial Intelligence in real-world applications, with openness to relocation.

Experience

Data Scientist - Machine Learning Engineer

02/2025 - Present

Centre for Research & Technology Hellas (CERTH)

Responsibilities include:

Project 1 - Enact: AI Forecaster For Cloud & Edge Telemetry Forecasting

- Designed an end-to-end forecasting pipeline for multivariate telemetry data (CPU, RAM, latency, energy).
- Developed time-series forecasting models in PyTorch, using both custom hybrid LSTM-Transformer architectures and pre-trained forecasting transformers such as Chronos.
- Deployed and orchestrated containerized inference services on Kubernetes using Docker, with scheduled jobs for automated predictions.
- Integrated MinIO object storage for managing telemetry data, models, and predictions.

Skills: Python, SQL, Pytorch, Kubernetes, Docker, Prometheus, Grafana, Databricks, CI/CD, Git, Gitlab/Github FastAPI, Flask, Scikit-learn.

Project 2 - Extra-Brain: Generative AI For Synthetic Data & Image Generation

- Designed convolutional architectures for image reconstruction and generation.
- Implemented VAEs, Conditional VAEs, and GANs for synthetic data generation and augmentation.
- Utilized pre-trained diffusion models for image synthesis and latent-space analysis.
- Designed, implemented, and deployed a data-preprocessing dashboard to production using a microservices architecture with nginx reverse-proxy routing and Docker Compose.
- Built an explainable workflow with local LLMs (Ollama) that identifies under-represented classes and directs vision models to generate matching samples, improving dataset balance.

Skills: Gans, AutoEncoders, LLMs, Ollama, Generative Pretrained models, Diffusion Models, Vision Transformers, Cvat, Roboflow, YOLO, Docker compose, Nginx, Microservices.

Education

MSc in Artificial Intelligence | Aristotle University of Thessaloniki

09/2025 – Present

- On-site program comprising two academic semesters of coursework and one semester dedicated to the Master's thesis. Focus areas include Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning & Reinforcement Agents, Intelligent Systems and Neuroscience inspired Machine Learning.

Electrical & Computer Engineer | Democritus University of Trace

10/2018 – 12/2024

- Electronics and Information Systems
- Integrated Master (300 ECTS)
- Diploma Thesis titled "Imperfection Detection from Photovoltaic-Panel Electroluminescence Images".

An automated system for defect detection and categorization of photovoltaic panels was developed using deep learning. Four datasets were created and used to train Custom Convolutional Neural Networks (CNNs). Furthermore, pretrained models such as VGG-16, VGG-19, ResNet-18, and ResNet-34 were also utilized through transfer learning. The results of these models were compared with the CNNs' outcomes, evaluating their accuracy and efficiency.

Skills

- C, C++
- Object-Oriented Programming (OOP)
- Data Structures
- SystemVerilog, VHDL
- Microsoft Excel, Powerpoint, Word

Certifications & Publications

- Kapetanidou, I. A., Kotsiopoulos, T., **Thanasoulis, G.**, Bizopoulos, P., Liatifis, A., Skoularikis, M., Nizamis, A., Sarigiannidis, P., & Votis, K., "A telemetry dataset on resource utilisation and power consumption in the edge-cloud continuum". *Data in Brief*, vol 66, June 2026, p. 112734. doi: 10.1016/j.dib.2026.112734.
- Python for Data Science and Machine Learning Bootcamp - *Udemy*
- The Complete Neural Networks Bootcamp: Theory, Applications – *Udemy*
- The Complete SQL Bootcamp: Go from zero to Hero - *Udemy*

Languages

- Greece – Native
- English – Full Professional Proficiency (C2)